SOFTWARE REQUIREMENTS SPECIFICATION · IEEE 830 STRUCTURE

Camera-Based Navigation System for Visually Impaired People

A camera on a cap streams to an Android phone. The phone detects obstacles, estimates distance, picks a safe direction, and speaks instructions through Bluetooth earphones.

Version	1.0 (draft for review)
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Guide	Prof. Avanti Patil
Supersedes	Ceiling-camera + BLE design (Phase-2 report)

1 Introduction

1.1 Purpose

This document specifies the requirements for a wearable navigation aid. It is written for the project team, the project guide and the evaluators. It is the reference for design, implementation and acceptance testing.

1.2 Scope

The system has two parts:

- **Cap unit.** A small camera board mounted on the front of a cap. It captures the scene in front of the user and streams it over Wi-Fi.
- **Android application.** It receives the frames and runs all AI processing on the phone: object detection, obstacle detection, distance estimation and path analysis. It then decides a direction and speaks it through the phone speaker or Bluetooth earphones.

The system works indoors and outdoors at walking speed, with no internet connection and no building infrastructure. It **supplements** the white cane and does not replace it. Turn-by-turn route navigation (for example “go to Room 204”) is out of scope for this version.

1.3 Definitions and abbreviations

TERM	MEANING
Cap unit	The camera board, battery and optional IMU mounted on the cap.

TERM	MEANING
Walking corridor	The strip of the image directly ahead of the user, about shoulder width (0.8 m) at ground level.
Obstacle	Any object in or near the walking corridor that the user could collide with or trip over.
Zone	One of three vertical sectors of the view: left, centre, right.
Pitch	The up/down tilt of the user's head, and therefore of the camera.
MJPEG	Motion JPEG. A video stream sent as a sequence of JPEG images over HTTP.
TFLite	TensorFlow Lite (LiteRT). The on-device inference runtime for Android.
YOLOv8n	The nano variant of the YOLOv8 object detector, already used in Phase 2.
TTS	Text-to-speech.
IMU	Inertial measurement unit (accelerometer and gyroscope), e.g. MPU6050.
Must / Should / Could	MoSCoW priority. *Must* is required for the demo. *Should* is expected but can be cut if time runs out. *Could* is an enhancement.

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Project Phase-2 report: *Bluetooth Low Energy and Surveillance Camera-based Navigation System for Visually Impaired People*, AITM Belagavi, 2025–26.
- Phase-2 source code: `blenav/` (in particular `guidance.py`, `egocentric.py`, `voice.py`) and `esp32_cam_ap/` firmware.
- Ultralytics YOLOv8 documentation (TFLite export). Android TextToSpeech API documentation.

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2 Overall description

2.1 Product perspective

This is the third iteration of the project. Phase 2 used a fixed ceiling camera to locate the user and BLE to identify them, with processing on a laptop. Its "first-person mode" showed that a camera carried by the user works as well. This version adopts that first-person design as the main product and moves all processing onto the phone, so the user carries the whole system.

The following Phase-2 components are reused in adapted form:

- **Navigation policy** (`guidance.decide()`, 40 unit tests): ported to Kotlin with the same priority levels.
- **Size-based distance model** (`egocentric.py`, 56 unit tests): object height and width tables, reused as-is.

- **Voice priority and cooldown rules** (`voice.py`): re-implemented on Android TTS.
- **ESP32-CAM access-point firmware** (`esp32_cam_ap.ino`): reused with the pin map changed for the new board.

2.2 Product functions

1. Capture the scene in front of the user from a cap-mounted camera.
2. Stream frames to the Android phone over a direct Wi-Fi link.
3. Detect known objects (people, chairs, tables, doors, vehicles, stairs and others).
4. Detect obstacles in the walking corridor, including ones the detector cannot name.
5. Estimate the distance to each obstacle in metres.
6. Analyse the path ahead and find the clearest direction.
7. Decide an instruction and a safety alert level.
8. Speak short instructions through the phone or Bluetooth earphones.

2.3 User classes

USER	DESCRIPTION	WHAT THEY NEED
Primary user	A blind or low-vision person. May not be able to see the phone screen at all.	Start, stop and use the system entirely by voice, touch and hardware buttons. Short, clear spoken instructions.
Helper	A family member, caregiver or instructor who sets the system up.	Pair the cap, set the user's height and the camera angle, and adjust voice settings.
Developer / evaluator	The project team and examiners.	A debug view showing detections, distances and decisions live, plus logs.

2.4 Operating environment

- **Phone:** Android 10 (API 29) or later, 4 GB RAM or more, a mid-range chipset (Snapdragon 6/7 series, Dimensity 700 or better).
- **Cap unit:** ESP32-S3 camera board with OV2640 sensor, 2.4 GHz Wi-Fi, powered by a 3.7 V LiPo battery.
- **Audio:** the phone speaker or any Bluetooth earphones (A2DP). Open-ear or bone-conduction earphones are recommended.
- **Surroundings:** indoor corridors, rooms and campus pathways, in daylight or normal indoor lighting.

2.5 Design and implementation constraints

- All AI processing runs **on the phone**. No cloud service and no laptop in the loop.
- The app is written in Kotlin with Android Studio. Models run on TensorFlow Lite with the GPU or NNAPI delegate where available.
- The cap unit must be light and small enough to wear comfortably (target ≤ 100 g including battery).

- All thresholds (stop distance, warning distance, cooldowns) are configurable, not hard-coded. This follows the Phase-2 rule.
- The system must work with no internet connection.

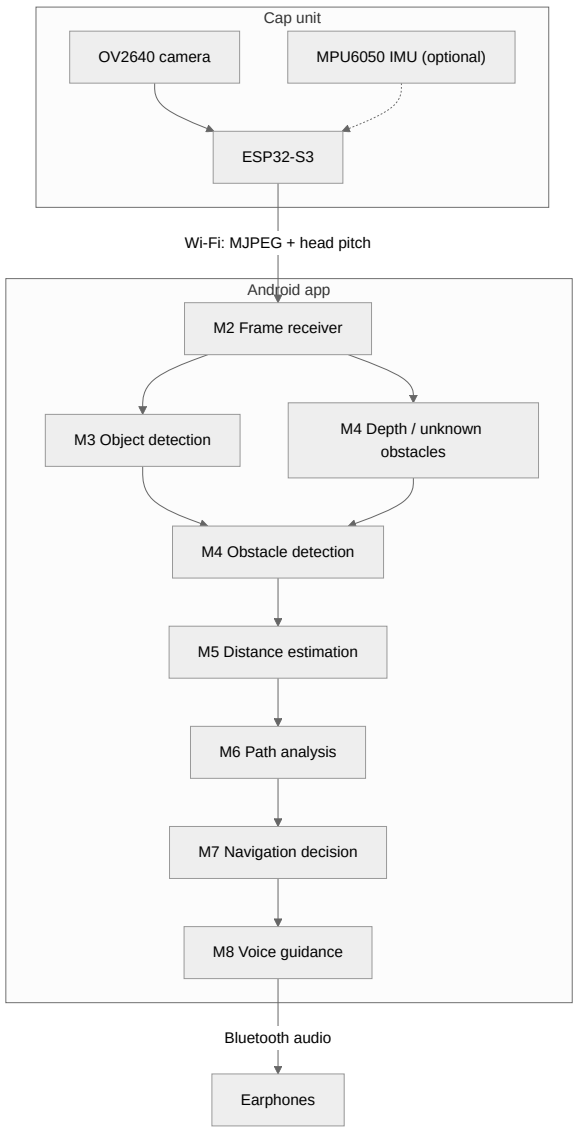
2.6 Assumptions and dependencies

- The user wears the cap with the camera facing forward and can hold their head roughly level while walking.
- Lighting is sufficient for a phone-grade camera. Night and very dark scenes are out of scope.
- The phone has an English TTS voice installed (available on all Google-certified phones).
- Stairs detection depends on a YOLO model fine-tuned on a stairs dataset, as planned in Phase 2.

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3 System architecture

The phone runs a four-stage pipeline. As in Phase 2, the frame receiver always keeps only the newest frame. Decisions run at a fixed 10 Hz whether detection is fast or slow.



3.1 Modules

ID	MODULE	RUNS ON	RESPONSIBILITY
M1	Cap camera unit	ESP32-S3	Capture frames, host the Wi-Fi hotspot, stream MJPEG, report head pitch.
M2	Frame receiver	Android	Connect to the cap, decode frames, drop stale frames, detect link loss.
M3	Object detection	Android	YOLOv8n (TFLite, INT8): class, box and confidence per object.
M4	Obstacle detection	Android	Decide which detections are obstacles. Find unrecognised obstacles from a depth map.
M5	Distance estimation	Android	Distance in metres from object size and from ground-plane geometry, combined.
M6	Path analysis	Android	Score the left, centre and right zones and pick the clearest.
M7	Navigation decision	Android	Priority policy that produces one instruction and an alert level.
M8	Voice guidance	Android	Speech queue with priorities, interruption and cooldown. Optional vibration.
M9	App shell and settings	Android	Accessible controls, setup screen, debug overlay, logs.

4 Functional requirements

Must needed for the demo **Should** expected, can be cut **Could** enhancement

4.1 M1 Cap camera unit

ID	REQUIREMENT	PRIORITY
FR-CAM-01	The cap unit shall capture colour frames at 320×240 (QVGA) or higher, at 10 frames per second or more.	Must
FR-CAM-02	The cap unit shall create its own Wi-Fi access point, so it works without college Wi-Fi.	Must
FR-CAM-03	The cap unit shall serve an MJPEG stream over HTTP (port 81, as in Phase 2).	Must
FR-CAM-04	The cap unit shall allow resolution and JPEG quality to be changed from the app.	Should

ID	REQUIREMENT	PRIORITY
FR-CAM-05	The cap unit shall read head pitch from an IMU and send it with the stream at 10 Hz or more.	Should
FR-CAM-06	The cap unit shall report its battery level to the app.	Could

4.2 M2 Frame receiver

ID	REQUIREMENT	PRIORITY
FR-RX-01	The app shall connect to the cap's Wi-Fi and bind its network traffic to that network, so Android does not route the stream over mobile data.	Must
FR-RX-02	The app shall keep only the newest frame and drop older ones, so processing never falls behind real time.	Must
FR-RX-03	If no frame arrives for 2 seconds, the app shall say "Camera disconnected. Please stop." and keep retrying.	Must
FR-RX-04	When the stream comes back, the app shall say "Camera connected" and resume guidance.	Should
FR-RX-05	The app shall accept the phone's own camera or a video file as a frame source for testing.	Should

4.3 M3 Object detection

ID	REQUIREMENT	PRIORITY
FR-DET-01	The app shall run YOLOv8n, exported to TFLite, on the phone and output class, bounding box and confidence per object.	Must
FR-DET-02	The detector shall recognise at least: person, chair, bench, table, door, bicycle, motorcycle, car, potted plant, pole/sign, and the other navigation-relevant classes from the Phase-2 list.	Must
FR-DET-03	The detector shall ignore detections below a configurable confidence threshold (default 0.40).	Must
FR-DET-04	The detector shall recognise stairs using a fine-tuned model.	Should
FR-DET-05	The detector shall use an INT8-quantised model with the GPU or NNAPI delegate when the phone supports it, and fall back to CPU otherwise.	Should

4.4 M4 Obstacle detection

ID	REQUIREMENT	PRIORITY
FR-OBS-01	The app shall treat any detected object whose box overlaps the walking corridor as an obstacle.	Must
FR-OBS-02	The app shall keep reporting an obstacle for a short hold period (default 0.6 s) after it is last seen. One missed frame shall never be treated as “path clear”.	Must
FR-OBS-03	The app shall ignore flat items that are not a collision risk (e.g. book, phone, keyboard), as in Phase 2.	Should
FR-OBS-04	The app shall run a lightweight monocular depth model (e.g. MiDaS small or Depth Anything V2 small, in TFLite) to find obstacles the detector cannot name, such as walls, pillars and boxes.	Should
FR-OBS-05	The app shall detect a drop or step down in the path ahead.	Could

4.5 M5 Distance estimation

ID	REQUIREMENT	PRIORITY
FR-DIST-01	The app shall estimate distance from apparent object size, using the Phase-2 table of real-world heights and widths and the camera’s field of view.	Must
FR-DIST-02	The app shall estimate distance for objects touching the floor from the bottom edge of the box, using camera height (user height minus about 0.1 m) and head pitch.	Should
FR-DIST-03	When both estimates exist, the app shall combine them and use the more conservative (nearer) value for safety alerts.	Should
FR-DIST-04	The app shall smooth distances over time per tracked object so the spoken distance does not jump between frames.	Should
FR-DIST-05	The helper shall be able to set the camera field of view and the user’s height in settings.	Must

4.6 M6 Path analysis

ID	REQUIREMENT	PRIORITY
FR-PATH-01	The app shall divide the view into left, centre and right zones and record the nearest obstacle distance in each.	Must
FR-PATH-02	When the centre zone is blocked, the app shall choose the side zone with the most clear distance. If both sides are blocked, it shall report no clear path.	Must

ID	REQUIREMENT	PRIORITY
FR-PATH-03	The app shall use the depth map to measure free space per zone, including unrecognised obstacles.	Should
FR-PATH-04	The app shall avoid switching direction on every frame (hysteresis), so the user does not hear “left, right, left”.	Must

4.7 M7 Navigation decision

The decision runs 10 times per second and returns one instruction using this priority order. It is adapted from the Phase-2 first-person thresholds.

LEVEL	TRIGGER	SPOKEN INSTRUCTION
P0 Stop	Obstacle ≤ 1.2 m in the centre zone, or camera lost	“Stop. Chair ahead.” / “Stop. Camera disconnected.”
P1 Hazard	Stairs or a drop ≤ 4.0 m ahead	“Caution. Stairs ahead.”
P2 Avoid	Obstacle ≤ 2.5 m in the centre zone	“Person ahead. Move left.”
P3 Blocked	Centre and both sides blocked	“Path blocked. Stop and turn.”
P4 Clear	Nothing within the warning distance	“Path clear.” (at most every 6 s)

ID	REQUIREMENT	PRIORITY
FR-NAV-01	The app shall implement the priority policy above as a pure function with unit tests, ported from `guidance.decide()`.	Must
FR-NAV-02	The app shall run the decision at a fixed 10 Hz, independent of detection speed.	Must
FR-NAV-03	All thresholds shall be read from a configuration file and editable in the settings screen.	Must
FR-NAV-04	The app shall not repeat the same instruction within a cooldown (default 3 s) unless its level rises.	Must
FR-NAV-05	On any internal failure the app shall fail safe: it shall say “Stop” rather than go silent.	Must

4.8 M8 Voice guidance

ID	REQUIREMENT	PRIORITY
FR-VOICE-01	The app shall speak instructions with Android TextToSpeech, with no internet connection.	Must
FR-VOICE-02	Audio shall play through connected Bluetooth earphones when present, and the phone speaker otherwise.	Must
FR-VOICE-03	A P0 instruction shall interrupt any speech in progress.	Must
FR-VOICE-04	Instructions shall be at most five words where possible, using the same wording every time.	Must
FR-VOICE-05	The user shall be able to change speech rate and volume.	Should
FR-VOICE-06	The phone shall vibrate on P0 alerts, in addition to speech.	Should
FR-VOICE-07	The app shall offer Kannada or Hindi instructions where the phone has that TTS voice installed.	Could

4.9 M9 App shell and settings

ID	REQUIREMENT	PRIORITY
FR-APP-01	The main screen shall have one full-screen Start/Stop control, usable with TalkBack.	Must
FR-APP-02	The app shall announce its state aloud: “Navigation started”, “Navigation stopped”, “Connecting to camera”.	Must
FR-APP-03	Navigation shall keep running with the screen off (foreground service).	Must
FR-APP-04	A settings screen for the helper shall hold: camera address, field of view, user height, distances, cooldowns, voice rate.	Must
FR-APP-05	A debug view shall show the live frame with boxes, distances, zone scores, current decision and FPS, for development and the demo.	Must
FR-APP-06	The user shall be able to start and stop navigation with a hardware volume-button press.	Should
FR-APP-07	The app shall keep a local log of decisions and timings for evaluation. It shall not save camera frames unless recording is switched on.	Should

5 External interface requirements

5.1 User interface

- The primary interface is **audio**. The screen is optional for the primary user.
- Every control has a TalkBack label. Touch targets are at least 48 × 48 dp.
- The debug view and settings are for sighted helpers and developers.

5.2 Hardware interfaces

LINK	INTERFACE	DETAILS
Camera → ESP32-S3	DVP parallel bus	OV2640 through the board's camera connector.
IMU → ESP32-S3	I²C	MPU6050 at address 0x68, sampled at 50 Hz, filtered to pitch.
Cap unit → phone	Wi-Fi 802.11n, 2.4 GHz	Cap is the access point. Stream at `http://192.168.4.1:81/stream`. Pitch at `/imu` (JSON).
Phone → earphones	Bluetooth A2DP	Handled by Android audio routing.
Battery → cap unit	3.7 V LiPo	Charged through the board's USB-C port.

5.3 Software interfaces

- Android SDK, minimum API 29, target API 35 or later. Kotlin.
- TensorFlow Lite (LiteRT) with GPU and NNAPI delegates.
- Android TextToSpeech, AudioManager and Vibrator services.
- Android ConnectivityManager, used to bind to the cap's Wi-Fi network.
- Arduino-ESP32 core with `esp_camera` on the cap unit.
- Ultralytics (Python, on the development PC only) to train, fine-tune and export models.

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6 Non-functional requirements

ID	REQUIREMENT	PRIORITY
NFR-PERF-01	The app shall process at least 8 frames per second on the target phone.	Must

ID	REQUIREMENT	PRIORITY
NFR-PERF-02	From the camera capturing an obstacle to the start of a P0 spoken alert shall take no more than 1.0 s (target 0.7 s).	Must
NFR-ACC-01	Distance estimates between 1 m and 4 m shall be within ±20% of a tape-measured distance for the supported object classes.	Should
NFR-ACC-02	In the test course, at least 90% of obstacles in the walking corridor shall produce a warning before the user is within 1.2 m.	Must
NFR-REL-01	Loss of the camera, model or voice engine shall produce a spoken warning within 2 s. The app shall never go silent while guidance is on.	Must
NFR-REL-02	The app shall run for 60 minutes without crashing or leaking memory.	Must
NFR-PWR-01	The cap unit shall run for at least 2 hours on one charge.	Should
NFR-PWR-02	The phone shall be able to navigate for at least 90 minutes from full charge.	Should
NFR-ERG-01	The cap unit shall weigh no more than 100 g and shall not become uncomfortably hot on the head.	Must
NFR-USE-01	A blind user shall be able to start guidance within 10 s of opening the app, without sighted help, after the first setup.	Must
NFR-PRIV-01	Frames shall be processed in memory and not uploaded. Nothing leaves the phone.	Must
NFR-SEC-01	The cap's Wi-Fi shall use WPA2 with a password that can be changed.	Should
NFR-MAINT-01	Decision, distance and path logic shall be pure functions with unit tests that run without hardware (as in Phase 2).	Must
NFR-SAFE-01	The app and all documentation shall state that the system supplements, and does not replace, the white cane.	Must

7 Acceptance tests

Run on a marked indoor course, about 15 m long, with tape-measured obstacle positions.

ID	SCENARIO	PASS CONDITION	COVERS
AT-01	Walk toward a chair placed in the path.	"Chair ahead. Move ..." at about 2.5 m, then "Stop" at about 1.2 m if the user keeps going.	FR-DET, FR-DIST, FR-NAV
AT-02	Chair on the left, clear on the right.	Instruction says "Move right" every time.	FR-PATH-01/02

ID	SCENARIO	PASS CONDITION	COVERS
AT-03	A person walks across the path.	Warning before they are within 1.2 m. No “Path clear” while they are in the corridor.	FR-OBS-02
AT-04	Approach a staircase.	“Caution. Stairs ahead.” at 3 m or more.	FR-DET-04
AT-05	Approach a plain wall or a cardboard box.	Warning from the depth model before 1.2 m.	FR-OBS-04
AT-06	Switch off the cap unit while walking.	“Camera disconnected. Please stop.” within 2 s.	FR-RX-03, NFR-REL-01
AT-07	Connect Bluetooth earphones mid-run.	Speech moves to the earphones with no restart.	FR-VOICE-02
AT-08	Measure latency with a flashing marker in view.	Capture-to-speech ≤ 1.0 s over 20 trials.	NFR-PERF-02
AT-09	Distance check: objects at 1, 2, 3 and 4 m.	Within ±20% for at least 80% of readings.	NFR-ACC-01
AT-10	A blindfolded tester starts the app and walks the course.	Starts without help. Completes the course with no collisions.	NFR-USE-01

8 Traceability to the stated requirements

GIVEN REQUIREMENT	MODULE	REQUIREMENT IDS
Camera mounted on cap	M1	FR-CAM-01…06, NFR-ERG-01
Captures the user’s surroundings	M1	FR-CAM-01, FR-CAM-03
Android mobile application	M2, M9	FR-RX-*, FR-APP-*
Receives camera frames and processes images	M2	FR-RX-01…05
AI-based image processing	M3–M6	FR-DET, FR-OBS, FR-DIST, FR-PATH
Object detection	M3	FR-DET-01…05
Obstacle detection	M4	FR-OBS-01…05
Distance estimation	M5	FR-DIST-01…05
Path analysis	M6	FR-PATH-01…04
Navigation decision module	M7	FR-NAV-01…05

GIVEN REQUIREMENT	MODULE	REQUIREMENT IDS
Determines direction and generates safety alerts	M6, M7	FR-PATH-02, FR-NAV-01, FR-NAV-05
Voice guidance	M8	FR-VOICE-01…07
Spoken instructions through phone or Bluetooth earphones	M8	FR-VOICE-02

9 Open issues and assumptions to confirm

This draft assumes the answers below. Change them here if the team decides otherwise.

QUESTION	ASSUMED IN THIS DRAFT
Where does image processing run?	On the Android phone (native Kotlin app, TFLite). No laptop.
Which camera goes on the cap?	An ESP32-S3 camera board (XIAO ESP32S3 Sense). The existing ESP32-CAM is a budget fallback.
What happens to BLE and the ceiling camera?	Removed from this version. Kept in the report as the previous design and as a comparison.
Minimum path analysis?	Three-zone (left/centre/right) analysis is required. Full depth-based free space is “Should”.
Language of instructions?	English required. Kannada/Hindi optional.

A Appendix A: Hardware to buy

Prices are approximate Indian online prices (Robu.in, Amazon.in and similar) and will vary by seller. Check stock before ordering.

A.1 Required

ITEM	WHY	QTY	APPROX. ₹
Seeed Studio XIAO ESP32S3 Sense (with OV2640 camera)	Thumb-sized camera board with 8 MB PSRAM and a faster chip than the ESP32-CAM. It has USB-C and a built-in LiPo charger, so no separate programmer or charging board is needed. Best fit for a cap.	1	1,400–1,800

ITEM	WHY	QTY	APPROX. ₹
3.7 V LiPo battery, 1000–1200 mAh, with JST connector	Powers the cap unit for 2–3 hours. Charged through the XIAO's USB-C port.	1	300–450
Small stick-on heatsink (aluminium, ~10 mm)	The XIAO Sense gets warm when streaming continuously. It sits against the user's head.	1–2	30–80
Mini slide switch	Power on/off without unplugging the battery.	2	20–40
Plain baseball cap with a stiff front panel	Mount for the camera. A stiff peak keeps the camera angle stable.	1	200–400
Enclosure: 3D-printed case or small ABS project box, plus Velcro and zip ties	Holds the board and battery. Must leave the camera clear and allow airflow.	1	200–500
Bluetooth earphones, **open-ear or bone-conduction** preferred	Bone-conduction leaves the ears open to traffic and voices, which matters for a blind user. Ordinary Bluetooth earbuds work for testing.	1	1,500–3,500 (bone) · 800–1,500 (earbuds)
Jumper wires, thin silicone wire, small perfboard, heat-shrink	Wiring the IMU and switch.	1 set	150–300

A.2 Recommended

ITEM	WHY	QTY	APPROX. ₹
MPU6050 IMU module (GY-521)	Measures head tilt. Needed for ground-plane distance (FR-DIST-02). Without it, distance relies on object size only.	1	120–250
Wide-angle OV2640 camera, 120°–160° lens (24-pin, fits XIAO)	Wider view catches obstacles to the sides. Only switch if the standard lens proves too narrow. Wide lenses distort, so the FOV setting must change too.	1	250–450
Spare LiPo battery	Swap during long test sessions and on demo day.	1	300–450

A.3 Optional

ITEM	WHY	QTY	APPROX. ₹
VL53L1X time-of-flight distance sensor (up to 4 m)	A true distance reading straight ahead. It covers plain walls that the camera may miss. Adds wiring and firmware work, so only if time allows.	1	600–1,000
ESP32-CAM-MB programmer board	Only if you use the existing AI-Thinker ESP32-CAM as a backup camera. The XIAO does not need it.	1	150–250

A.4 Do not buy

- **Raspberry Pi or Jetson Nano.** All processing runs on the phone, so a separate computer adds weight, cost and power with no benefit.
- **A new phone.** Use a team member's phone that meets section 2.4.
- **HC-SR04 ultrasonic sensors.** They are 5 V, bulky on a cap, and noisy. Use the VL53L1X if you want a range sensor.

A.5 Budget summary

KIT	CONTENTS	APPROX. ₹
Minimum	A.1 with ordinary Bluetooth earbuds	3,100–5,100
Recommended	A.1 with bone-conduction earphones, plus A.2	4,500–8,200
Recommended + optional	Above, plus VL53L1X	5,100–9,200

Budget route with the existing ESP32-CAM

If you already own the AI-Thinker ESP32-CAM from Phase 2, you can skip the XIAO. You then need an ESP32-CAM-MB programmer, a TP4056 charger board and an MT3608 5 V boost board (or a small power bank in a pocket with a cable to the cap). Expect a bulkier, heavier cap unit and a lower frame rate. The firmware in `esp32_cam_ap/` runs on it unchanged. For the XIAO, only the camera pin map in that firmware needs changing.